Reg. No:

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccreditedby NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - MAY'2025

COMMON TO B.E – CSE & M.TECH - CSE

20CS211 - ALGORITHMS AND COMPLEXITY- II

Answer ALL questions

Duration: 3 Hours **Maximum: 100 Marks**

PART - A **(7x 1 = 7 Marks)**

1. **CO2** **Choose the correct algorithm that makes use of the Transform and Conquer paradigm?**
   a) Dijkstra's Algorithm b) Floyd-Warshall Algorithm
   c) Heap Sort d) Breadth-First Search

2. **CO2** **Identify in which of the following scenarios is "problem transformation" used in the context of Transform and Conquer?**
   a) Using AVL trees for faster search
   b) Converting a graph problem into a matrix problem
   c) Preprocessing strings for KMP algorithm d) Building a prefix tree for a dictionary

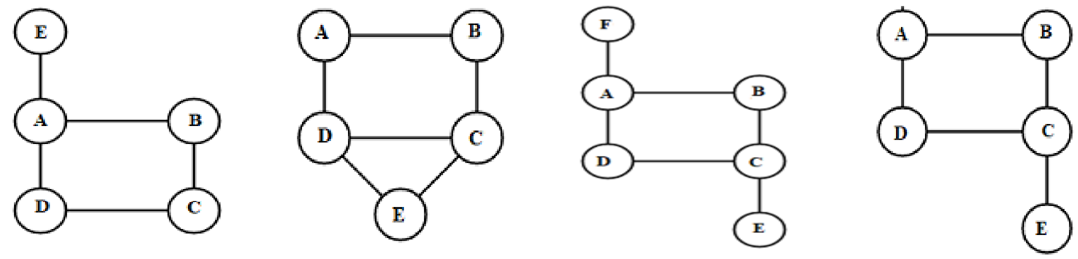
3. **CO3** **Choose the Correct statement for the assumption P!= NP.**
   a) NP-complete = NP b) NP-complete ∩P=Φ
   c) NP-hard = NP d) P = NP-complete

4. **CO2** **Select the correct edge that can be used to identify an articulation points of a graph.**
   a) Forward b) Tree c) Backedge d) Cross

5. **CO2**
```
int mystery(int n) {
   if (n <= 1) return 1;
   return 1 + mystery(n - 1);}
```
   This function demonstrates:
   a) Divide and Conquer b) Increase and Conquer
   c) Decrease and Conquer (by constant) d) Backtracking

6. **CO2** **Apply the backtracking technique to the given graphs and identify which graph allows the construction of a Hamiltonian circuit.**

a) b) c) d)



7. **CO2** **Choose the correct order:**

a) Dijkstra's Algorithm: 1) MST

b) Floyd's Algorithm: 2) Transitive Closure

c) Warshall's Algorithm: 3) Single Source Shortest Path

d) Kruskal's Algorithm: 4) All Pair Shortest Algorithm

a) abcd:1234 b) abcd:2143 c) abcd:3421 d) abcd:4312

Fill in the blanks: **(3x 1 = 3 Marks)**

8. **CO2** A node in a state space tree is said to be ______, if it corresponds to a partially constructed solution that may still lead to a complete solution.

9. **CO3** No polynomial time has yet been discovered for an ______ problem.

10. **CO1** ______ Tree is used for encoding and decoding.

PART - B **(5x 3 = 15 Marks)**

11. **CO2** Apply Horner's rule to evaluate the polynomial p(x) = 3x4 − x3 + 2x + 5 at x = −2.

12. **CO1** Define topological sorting.

13. **CO2** Vertices: A, B, C Edges: A → B (3), B → C (4), A → C (10)

Apply Floyd's algorithm to find the shortest path from vertex A to all other vertices.

14. **CO2** Show the steps in solving to the placing 4 queens on a 4x4 chessboard.

15. **CO3** State the purpose of polynomial time reduction in Complexity analysis.

PART - C **(5 x 15 = 75 Marks)**

Q.No: 16 Compulsory Question
Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO2** i) Apply heap sort using array representation for the list 2,9,7,6,5,8 in Max heap order. (8)

CO1 ii) Construct a top-down 2-3-4 tree by inserting the following list of keys in the initially empty tree: 10, 6, 15, 31, 20, 27, 50, 44, 18. (7)

17. **CO1** i) You are working as a software engineer for a logistics company that manages dynamic delivery routes using a search system backed by self-balancing binary search trees. To maintain fast search, insert, and delete operations, your team has chosen to use an AVL Tree for managing route priority keys. Your manager provides you with a new sequence of delivery route priorities that need to be inserted into the AVL Tree to maintain balance and ensure optimal performance. The sequence is :5, 6, 8,3,2,4,7.Illustrate the structure of the AVL Tree after each insertion and any rotations (single or double) performed during the process. (8)

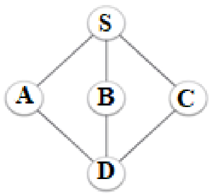
CO2 ii) Apply the Russian Peasant Multiplication algorithm to multiply two integers and the Interpolation Search algorithm to locate a target value in a sorted array with uniformly distributed elements (7)

18. **CO2** i) Construct a Huffman code for the following data: (8)

Character	A	B	C	D	-
Probability	0.35	0.1	0.2	0.2	0.15

Find the code word of each character by constructing tree.

CO1 ii) Apply the Breadth First Traversal algorithm for the given graph and give the BFS traversal ordering. (7)



19. **CO2** Predict the most valuable items to be shipped in a limited-capacity drone for a mountain expedition. Each item has a specific weight and value. However, due to the drone's weight limit, not all items can be included. (15)

To ensure maximum utility for the explorers, you need to pack the drone with a combination of items that maximizes total value without exceeding the drone's weight capacity. Because there are many items to choose from, and time is critical, you decide to use the Branch-and-Bound algorithm to solve this efficiently.

Drone's weight Capacity is 10.

ITEM	WEIGHT	VALUE
1	4	Rs.40
2	7	Rs.42
3	5	Rs.25
4	3	Rs.12

20. **CO2** Given a set of keys (vertices) along with their corresponding search probabilities. Construct an Optimal Binary Search Tree that minimizes the expected search cost. (15)

Keys: [A, B, C, D]
Probabilities: [A:0.2,B:0.5,C:0.3,D:0.1]

21. **CO3** The Boolean formula given as, (15)

$\mathbf{\Phi = C_1 \wedge C_2 \wedge C_3}$,
where:

- $C_1 = (x_1 \vee$ ___ $\vee$ ___$)$,
- $C_2 = ($___ $\vee x_2 \vee x_3)$,
- $C_3 = (x_1 \vee x_2 \vee x_3)$
- Construct a graph based on the 3-CNF-SAT formula where each literal in a clause is represented as a vertex, and edges are added between non-conflicting literals from different clauses.
- Analyze the number of cliques of size 3 (one literal per clause) that satisfy the formula Φ and reduce the clique problem into an equivalent Vertex Cover problem using a suitable graph transformation.
- Explain each step involved in the transformation and interpretation.

22. **CO3** A delivery company needs to determine an efficient route for a truck that must visit several cities exactly once and return to the starting point. Due to limited time, the company opts for approximate algorithms rather than exact methods. Apply the Nearest Neighbor algorithm, Minimum Spanning tree based Algorithm and the Multifragment heuristic algorithm to solve the routing problem for your own example with set of cities and distances. Illustrate each step of both algorithms clearly, and compare their efficiency and limitations in terms of solution quality and computational effort. (15)

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